

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech

Branch : Mechanical Engineering /Mechanical

Semester : III

Engineering (Sandwich) /Mechanical and Automation

Subject Code & Name: BTMC303 Thermodynamics

Max Marks: 60

Date:10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	A system in which energy can crosses the boundary of the system but the mass cannot is known as	1/CO- 1	1
	a. open system b. Closed system c. Isolated System d. None of These		
2	Which of the following is an intensive property of a thermodynamic system?	2/CO- 1	1
	a. mass 1/CO- 1 c. enthalpy d. density		
3	When two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other. This statement is called _____ of thermodynamics	2/CO- 1	1
	a. Zeroth Law b. First law c. second law d. third law		
4	First law of thermodynamics deals with _____	2/CO- 2	1
	a.Conservation of mass 1/CO- 1 c.Conservation of energy d.Conservation of pressure		
5	An increase in enthalpy leads to an increase in _____	1/CO- 2	1
	a. Increase in pressure b. Increase in internal energy c. Increase in volume d. Increase in mass		
6	SI unit of enthalpy is _____	2/CO- 3	1
	a. Joule/kg 1/CO- 1 c. Joule/K d. K/kg		
7	In Carnot cycle the heat receive at.... Temperature.	2/CO- 3	1
	a. Lower 1/CO- 1 c. constant d. None of the above		

8	Gases of the same volume contain the same number of molecules. This is the description of				2/CO- 4	1
	a. Boyle's law	b. Dalton's law	c. Charles' law	d. Avogadro's law		
9	The throttling process is _____ process.				2/CO- 3	1
	a. reversible	b. irreversible	c. reversible or irreversible	d. None of these		
10	Which of the following is not true about ideal gas molecules?				3/CO- 4	1
	a. They have negligible size	b. They do not have attractive forces	C .They do not apply pressure	d. They move in random motion		
11	The value of pressure of water at a critical point is in bar...				4/CO- 5	1
	a. 223.5 bar	b. 212.25 bar	c. 235.5 bar	d. 221.2 bar		
12	With the decrease in pressure the latent heat of steam is....				2/CO- 5	1
	a. Decreases	b. increases	c. Constant	d. Increases or decreases		
Q. 2	Solve the following.					12
A)	Explain Intensive and Extensive properties with suitable examples.				2/CO-1	6
B)	Explain 'Thermodynamic equilibrium' of a system and state its significance.				2/CO-1	6
Q.3	Solve the following.					12
A)	Explain the first law of thermodynamics for (i) closed system undergoing a cycle (ii) closed system a change of state.				2/CO-2	6
B)	A gas undergoes a thermodynamic cycle consisting of the following processes: (i) Process 1-2: Constant pressure $p = 1.4$ bar, $V_1 = 0.028$ m ³ , $W_{1-2} = 10.5$ KJ. (ii) Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$ (iii) Process 3-1: Constant volume $U_1 - U_3 = -26.4$ KJ. There are no significant changes in KE and PE. (a) Sketch the cycle on p-V diagram. (b) Calculate the net work for the cycle in KJ. (c) Calculate the net heat transfer for the process 1-2 (d) Show that $\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$				3/CO-2	6
Q. 4	Solve Any Two of the following.					12
A)	Define the term COP as applied to a refrigerator and heat pump. Show: $\text{COP}_{\text{HP}} = \text{COP}_{\text{REF}} + 1$				4/CO-3	6

B)	A cyclic heat engine operates between a source temperature of 1000°C and sink temp of 40°C. Find the least rate of heat rejection per KW net output of the engine?	4/CO-3	6
C)	Show that Entropy changes in reversible process is Zero	4/CO-3	6
Q.5	Solve Any Two of the following.		12
A)	State 'Avagadro's Hypothesis' and from it determine the value of Universal gas constant.	1/CO-4	6
B)	Define specific heats. Derive relationship between C_p and C_v .	2/CO-4	6
C)	Derive expression for work done during polytropic process for a non-flow system	4/CO-4	6
Q. 6	Solve Any Two of the following.		12
A)	What is pure substance? Give suitable example. What is saturation pressure and saturation temperature? Explain it on T-S diagram.	2/CO-5	6
B)	A vessel having a capacity of 0.04 m ³ contains a mixture of saturated water and saturated steam at a temperature of 250°C. The mass of steam present is 9 kg. Find the following i. The Pressure ii. The mass iii. The specific enthalpy iv. the specific volume v. the specific entropy vi. The specific internal volume	4/CO-5	6
C)	Draw a P-T (Pressure-Temperature) diagram of Pure substance (Water) and explain sublimation curve and Fusion curve	4/CO-5	6
*** End ***			